🫁 Medications for the Upper Respiratory System

Drug Class & Subclass	Prototype	Mechanism of Action	Key Side Effects / Adverse Effects	Nursing Interventions & Patient Teaching
First-Generation Antihistamine	**Diphenhydramine**	Competitively blocks H1 histamine receptors; crosses the blood-brain barrier.	• Severe sedation, drowsiness • Anticholinergic effects (dry mouth, blurred vision, urinary retention, constipation).	• **Implement fall precautions**due to severe sedation. • Teach patient to avoid driving and alcohol/CNS depressants. • Manage dry mouth with sugarless gum or hard candy.
Second-Generation Antihistamine	**Cetirizine**	Competitively blocks peripheral H1 histamine receptors; minimal crossing of blood-brain barrier.	• Minimal to no sedation • Mild anticholinergic effects.	• Prefers over first-generation for daytime allergy maintenance. • Advise patient that it is much less likely to cause drowsiness.
Opioid Antitussive	**Codeine**	Suppresses the cough reflex through direct action on the cough center in the medulla.	• **Respiratory depression** • Sedation, lightheadedness • Constipation.	• Monitor respiratory rate closely; hold if RR <12. • Advise changing positions slowly to prevent orthostatic hypotension. • Increase fluid and fiber intake to avoid constipation.
Nonopioid Antitussive	**Dextromethorphan**	Suppresses the cough reflex by direct action on the medullary cough center; lacks analgesic properties.	• Dizziness, drowsiness • Gastrointestinal upset/nausea.	• Avoid eating or drinking for 30 minutes after taking syrup forms. • Advise patient to notify provider if a high fever or persistent cough lasts >1 week.

🫁 Medications for the Lower Respiratory System

Drug Class & Subclass	Prototype	Mechanism of Action	Key Side Effects / Adverse Effects	Nursing Interventions & Patient Teaching
Short-Acting Beta$_2$ Agonist (SABA)	**Albuterol**	Stimulates beta$_2$-adrenergic receptors in airway smooth muscle, causing rapid bronchodilation.	• **Tachycardia, palpitations** • Tremors, nervousness, anxiety.	• **Used as a rescue inhaler**during acute bronchospasm episodes. • Monitor heart rate prior to and during therapy. • Teach correct Metered-Dose Inhaler (MDI) technique.
Long-Acting Beta$_2$ Agonist (LABA)	**Salmeterol**	Prolonged stimulation of beta$_2$-adrenergic receptors for long-term bronchodilation.	• Headache, nervousness • Tachycardia, muscle cramps.	• **Never use for acute asthma attacks**; it is for maintenance therapy only. • Often combined with an inhaled corticosteroid.
Short-Acting Muscarinic Antagonist (SAMA)	**Ipratropium**	Blocks acetylcholine receptors in bronchial smooth muscle, inhibiting bronchoconstriction.	• Dry mouth, throat irritation • Hoarseness, bitter taste.	• Often combined with albuterol (e.g., DuoNeb) for synergistic bronchodilation. • Rinse mouth after use to eliminate bitter taste.
Long-Acting Muscarinic Antagonist (LAMA)	**Tiotropium**	Long-acting blockade of muscarinic receptors in airways for maintenance COPD/asthma management.	• Dry mouth • Urinary retention, constipation.	• Administered daily via dry powder inhaler (DPI); **do not swallow the capsules**. • Monitor urinary output in patients with benign prostatic hyperplasia (BPH).
Inhaled Corticosteroid	**Fluticasone**	Suppresses inflammation by inhibiting immune response and reducing airway mucosal edema.	• **Oral candidiasis (thrush)** • Hoarseness (dysphonia), dry cough.	• **Instruct patient to rinse and spit** after every use to prevent thrush. • Administer *after*bronchodilators if taking both concurrently to optimize lung penetration.
Leukotriene Modifier	**Montelukast**	Blocks leukotriene receptors, reducing airway inflammation, bronchoconstriction, and mucus production.	• **Neuropsychiatric events** (suicidal ideation, vivid dreams) • Headache.	• Monitor patient closely for sudden changes in behavior, mood, or suicidal thoughts. • Administer orally once daily in the evening for asthma protection.

🧠 Medications to Treat Neuromuscular & Seizure Disorders

Drug Class & Subclass	Prototype	Mechanism of Action	Key Side Effects / Adverse Effects	Nursing Interventions & Patient Teaching
GABA Analog	**Gabapentin**	Structurally resembles GABA; modulates voltage-gated calcium channels to decrease excitatory neurotransmitter release.	• Somnolence, dizziness, fatigue • Peripheral edema.	• Advise patient not to abruptly discontinue due to withdrawal seizure risks. • Monitor for worsening depression or suicidal behavior.
Dopaminergic Agent	**Carbidopa & Levodopa**	Levodopa converts to dopamine in the brain; Carbidopa blocks peripheral breakdown of levodopa so it can cross the blood-brain barrier.	• **Orthostatic hypotension** • Dyskinesias (involuntary movements) • Dark-colored sweat/urine (benign).	• Teach patient to avoid high-protein meals, as protein reduces drug absorption. • Monitor for the "wearing-off" effect at the end of dosing cycles.
Sodium Channel Inhibitor	**Phenytoin**	Inhibits sodium channels, stabilizing neuronal membranes to suppress seizure activity.	• **Gingival hyperplasia** (gum overgrowth) • Nystagmus, ataxia, sedation • Teratogenicity.	• **Monitor therapeutic plasma concentration levels as ordered by the provider**; regular blood draws required. • Teach strict, meticulous oral hygiene and frequent dental cleanings. • Monitor for severe skin rashes (Stevens-Johnson syndrome).
GABA Augmenting Agent	**Valproic Acid**	Increases GABA availability, blocks sodium channels, and suppresses calcium influx.	• **Hepatotoxicity**(liver failure) • Pancreatitis • Thrombocytopenia.	• Monitor liver function tests (LFTs), amylase/lipase, and CBC (platelets). • Instruct patient to report immediate abdominal pain, jaundice, or bruising.
Benzodiazepine	**Lorazepam**	Binds to benzodiazepine receptors, enhancing the inhibitory effects of GABA in the CNS.	• **CNS depression/ sedation** • Respiratory depression • Tolerance and physical dependence.	• **Used for status epilepticus management as prescribed**. • Have the reversal agent, **flumazenil**, readily available. • Avoid abrupt discontinuation to prevent life-threatening withdrawal.

🧠 Medications to Treat Neurocognitive Disorders

Drug Class & Subclass	Prototype	Mechanism of Action	Key Side Effects / Adverse Effects	Nursing Interventions & Patient Teaching
Methylphenidate	**Methylphenidate**	Blocks norepinephrine and dopamine reuptake, increasing synaptic levels in the CNS.	• **Insomnia, weight loss/anorexia** • Tachycardia, hypertension.	• Administer earlier in the day as prescribed to prevent insomnia. • Monitor height, weight, and growth velocity in pediatric patients.
Acetylcholinesterase Inhibitor	**Donepezil**	Inhibits acetylcholinesterase, increasing acetylcholine levels available at cerebral synapses.	• Bradycardia, syncope • Nausea, vomiting, diarrhea.	• Administer at bedtime with or without food as prescribed. • Implement fall safety protocols due to potential bradycardia/ fainting risks.

🧠 Antidepressants & Anxiolytics

Drug Class & Subclass	Prototype	Mechanism of Action	Key Side Effects / Adverse Effects	Nursing Interventions & Patient Teaching
SSRI	**Fluoxetine**	Selectively inhibits synaptic serotonin reuptake, increasing serotonin levels.	• **Serotonin Syndrome** • Sexual dysfunction, insomnia • Weight gain.	• Monitor for **Serotonin Syndrome** (fever, tremors, hyperreflexia, agitation). • May take several weeks to achieve therapeutic antidepressant benefit. • Monitor closely for increased suicide risk in the first few weeks.
SNRI	**Venlafaxine**	Inhibits the reuptake of both serotonin and norepinephrine in the CNS.	• **Hypertension** • Insomnia, anxiety, nausea.	• Monitor blood pressure regularly before and during treatment. • Do not stop abruptly; discontinuation syndrome can cause severe dizziness and parasthesia.
NDRI	**Bupropion**	Inhibits the reuptake of norepinephrine and dopamine.	• **Seizures**(lowers seizure threshold) • Weight loss, insomnia, agitation.	• Contraindicated in patients with seizure disorders, anorexia, or bulimia. • Often preferred because it lacks sexual side effects and does not cause weight gain.
Tricyclic Antidepressant (TCA)	**Amitriptyline**	Blocks reuptake of serotonin and norepinephrine; also blocks histaminic, muscarinic, and alpha-adrenergic receptors.	• **Cardiotoxicity / Arrhythmias** • Severe anticholinergic effects • Sedation, orthostatic hypotension.	• Highly toxic in overdose; obtain a baseline EKG. • Administer at bedtime due to profound sedative properties.
MAOI	**Selegiline**	Inhibits monoamine oxidase, preventing the breakdown of dopamine, serotonin, and norepinephrine.	• **Hypertensive Crisis** (when combined with tyramine) • Localized skin irritation (transdermal patch).	• **Enforce a strict low-tyramine diet**(avoid aged cheeses, cured meats, red wine, beer). • Require an appropriate clinical washout period before starting any other antidepressant class.
5-HT$_1$A Partial Agonist	**Buspirone**	Binds to serotonin and dopamine receptors; does not cause generalized CNS depression.	• Dizziness, nausea • Headache.	• Used for generalized anxiety; **not for acute anxiety attacks** (requires ongoing use for clinical effect). • Lacks abuse potential, tolerance, or dependence; does not cause sedation.

🧠 Medications for Bipolar Disorders & Psychosis

Drug Class & Subclass	Prototype	Mechanism of Action	Key Side Effects / Adverse Effects	Nursing Interventions & Patient Teaching
Mood Stabilizer	**Lithium**	Alters ion transport in muscle and nerve cells; influences serotonin and norepinephrine reuptake.	• **Lithium Toxicity** • Fine hand tremors, polyuria • Hypothyroidism, renal impairment.	• **Monitor therapeutic serum concentration levels as prescribed**. • **Maintain adequate fluid intake and stable sodium balance** (low sodium triggers toxicity). • Contraindicated with NSAIDs and diuretics (they increase lithium levels).
First-Generation Antipsychotic (Typical)	**Chlorpromazine**	Blocks dopamine D2 receptors in the brain; possesses high anticholinergic properties.	• **Extrapyramidal Symptoms (EPS)** • **Neuroleptic Malignant Syndrome (NMS)** • Prolonged QT interval.	• Monitor for EPS (acute dystonia, akathisia, parkinsonism, tardive dyskinesia). • Watch for **NMS**(high fever, severe muscle rigidity, altered mental status, autonomic instability).
Second-Generation Antipsychotic (Atypical)	**Risperidone**	Blocks both serotonin 5-HT2 and dopamine D2 receptors.	• **Metabolic Syndrome** (weight gain, dyslipidemia, hyperglycemia) • Mild EPS (lower risk than typicals) • Hyperprolactinemia.	• Monitor fasting blood glucose, lipid panels, and baseline/ periodic weight. • Teach patients about healthy lifestyle modifications to mitigate metabolic effects.